

Ex.No: 3**Date:25/8/25**

CUSTOMIZED PING COMMAND TO TEST THE SERVER CONNECTIVITY

Aim:

To develop a customized ping command to test the server connectivity.

Introduction:

Ping:

The ping command sends ICMP echo request packets to a host and waits for echo replies. It measures reachability and reports RTT statistics (min, max, avg, packet loss).

Algorithm:

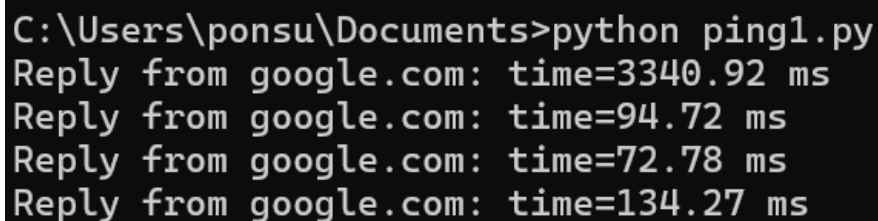
1. Set host, port, and number of pings.
2. Create an empty list to store RTT values.
3. Repeat for the given number of pings.
4. Start a timer.
5. Attempt to connect to host on the given port.
6. Stop the timer if connection is successful.
7. Calculate RTT and save it in the list.
8. If connection fails, print timeout.
9. After all attempts, find minimum, maximum, and average RTT.
10. Display the results.

Program Code:

Customized ping command to test the server connectivity:

```
import socket
import time
host = "google.com"
port = 80
count = 4
for i in range(count):
    try:
        s = socket.socket()
        start = time.time()
        s.connect((host, port))
        end = time.time()
        s.close()
        print(f'Reply from {host}: time={((end-start)*1000:.2f)} ms')
    except Exception:
        print("Request timed out")
```

Output:



```
C:\Users\ponsu\Documents>python ping1.py
Reply from google.com: time=3340.92 ms
Reply from google.com: time=94.72 ms
Reply from google.com: time=72.78 ms
Reply from google.com: time=134.27 ms
```

Program Code:

Customized Ping Program to Measure Min, Max, and Average RTT:

```
import socket, time

host = "google.com"

port = 80

count = 4

times = []

for i in range(count):

    try:

        s = socket.socket()

        start = time.time()

        s.connect((host, port))

        end = time.time()

        s.close()

        rtt = (end - start) * 1000

        times.append(rtt)

        print(f'Reply from {host}: time={rtt:.2f} ms')

    except:

        print("Request timed out")

if times:

    print("\nMin RTT =", min(times), "ms")

    print("Max RTT =", max(times), "ms")

    print("Avg RTT =", sum(times)/len(times), "ms")
```

Output:

```
C:\Users\ponsu\Documents>python ping2.py
Reply from google.com: time=179.05 ms
Reply from google.com: time=95.55 ms
Reply from google.com: time=102.88 ms
Reply from google.com: time=99.63 ms

Min RTT = 95.54910659790039 ms
Max RTT = 179.0475845336914 ms
Avg RTT = 119.2777156829834 ms
```

Result:

Thus the server connectivity is tested using ping command.